

Regression, Chi-Square & the Analyst Toolkit

Today's Big Question: Now that we know *if* two things move together (correlation) and *whether* a difference is real (hypothesis testing), how do we actually **predict** one variable from another, and how do we test relationships between **categories** instead of numbers?

Why This Matters

This is the capstone day. Regression is the single most-used technique in data analytics — it turns correlation into an actual predictive equation. Chi-square testing extends hypothesis testing to categorical data (yes/no, which group, which option) which makes up a huge share of real business data. We'll close by connecting everything this week to the actual day-to-day tools — SQL and dashboards — that turn these concepts into a real job.

By the End of Today, You Will Be Able To:

- Fit a simple linear regression line and explain slope and intercept in plain English
- Interpret R-squared and know what it does and doesn't tell you
- Run a chi-square test of independence on categorical data
- Know when to use a t-test vs. a chi-square test vs. regression
- Connect this week's concepts to SQL queries and dashboard reporting
- Walk away with a full mental map of the week: probability → correlation → testing → regression

Real-World Use Case: A real-estate analyst wants to predict house price from square footage (regression), and separately wants to know if buyer preference for "modern" vs. "traditional" style depends on the buyer's age group (chi-square). Both are everyday analyst tasks built on this week's foundation.

How to Introduce This Topic

Opening Line for the Class

Try this: "On Wednesday we said correlation tells you two things move together. But what if I told you exactly how much one moves when the other moves? What if I could give you an equation to predict the second variable just from the first?" That equation is regression — the natural next step after correlation.

Tie Back to the Whole Week

Day	Topic	What It Gave Us
Tuesday	Probability & Normal Distribution	How to think about uncertainty and what 'typical' looks like
Wednesday	Correlation & Covariance	Whether two variables move together, and how strongly
Thursday	Hypothesis Testing & P-Values	Whether an observed difference is real or just chance
Today	Regression, Chi-Square & Toolkit	How to predict one variable from another, and test category data

The Core Idea in One Sentence

Regression draws the best straight line through your data so you can predict Y from X. Chi-square asks whether two categorical variables (like gender and product preference) are related, the same way correlation asks that question for numeric variables.

PART 1

Simple Linear Regression — Predicting Y from X

Definition: Linear regression finds the straight line that best fits your data, so you can predict a numeric outcome (Y) from a numeric input (X).

The Equation Everyone Needs to Recognize

$Y = mX + b$ (or written in stats notation: $Y = \beta_0 + \beta_1 X + \varepsilon$)

- **Y** — the outcome you're predicting (e.g., house price)
- **X** — the input you're predicting from (e.g., square footage)
- **m / β_1 (slope):** for every 1-unit increase in X, Y changes by this much
- **b / β_0 (intercept):** the predicted value of Y when X = 0
- **ε (error term):** the part of Y that the line doesn't explain — real data never sits perfectly on a line

Worked Example: House Price vs. Square Footage

Square Footage (X)	Price in Rs Lakhs (Y)
1000	50
1200	58
1500	70
1800	82
2000	90

Talking point: ask the class to eyeball this and guess the slope before you calculate it. Roughly, every extra 200 sq ft adds about Rs 8 lakhs — so the slope is around 0.04 lakhs per sq ft. This primes their intuition before the formal calculation.

Regression in Python

```
import numpy as np
from scipy import stats

sqft = [1000, 1200, 1500, 1800, 2000]
price = [50, 58, 70, 82, 90] # Rs lakhs

slope, intercept, r_value, p_value, std_err = stats.linregress(sqft, price)

print(f"Slope (m): {slope:.4f}")
print(f"Intercept (b): {intercept:.2f}")
print(f"R-squared: {r_value**2:.3f}")
print(f"P-value: {p_value:.4f}")

# Predict the price of a 1600 sq ft house
predicted_price = slope * 1600 + intercept
print(f"Predicted price for 1600 sqft: Rs {predicted_price:.1f} lakhs")
```

How to Read the Output

- **Slope:** the predicted Rs-lakh change in price for every extra square foot
- **Intercept:** the theoretical price at 0 square feet — often not meaningful on its own, just needed to anchor the line
- **R-squared:** covered in detail next page — how well the line fits the data
- **P-value:** tests whether the slope is significantly different from zero (i.e., is there really a relationship, using the same hypothesis-testing logic from yesterday)

Connecting the dots: notice the p-value here is literally a hypothesis test on the slope — H_0 : slope = 0 (no relationship), H_1 : slope $\neq$ 0. This is yesterday's entire framework reused for regression. Point this out explicitly to tie the week together.

PART 2

R-Squared — How Good Is the Fit?

R-squared (R^2) tells you what percentage of the variation in Y is explained by X. It ranges from 0 to 1 (or 0% to 100%).

Interpretation Scale

R-Squared	Interpretation
0.90 – 1.00	Excellent fit — X explains almost all the variation in Y
0.70 – 0.90	Good fit — X explains most of the variation
0.50 – 0.70	Moderate fit — X explains roughly half the variation
0.30 – 0.50	Weak fit — X explains some, but a lot is unexplained
Below 0.30	Poor fit — X barely helps predict Y

The Direct Link to Wednesday

Here's a fact worth pointing out explicitly: **R-squared is just the correlation coefficient (r) squared**, for simple linear regression with one predictor. If Wednesday's correlation between marketing spend and revenue was $r = 0.99$, then $R\text{-squared} = 0.98$ — meaning marketing spend explains 98% of the variation in revenue in that dataset. This single fact often makes regression suddenly click for students who already understood correlation.

Important caveat: high R-squared does not mean your model is useful for every purpose, and it still says nothing about causation — the correlation-is-not-causation warning from Wednesday applies fully here too.

Regression Pitfalls to Watch For

Common Traps

- **Extrapolation:** never trust predictions far outside the range of your original data. A line fit on 1000–2000 sq ft homes should not be used to predict a 10,000 sq ft mansion
- **Outliers:** a single extreme data point can drag the whole regression line off course — always plot your data before trusting the numbers
- **Non-linear relationships:** if the true relationship curves, a straight line will systematically mis-predict at the high and low ends — always look at a scatterplot first
- **Confusing correlation strength with causation:** a strong, significant regression still doesn't prove X causes Y — the firefighters/fire-damage trap from Thursday applies here too

Talking Point: Always Plot Before You Model

Make this a hard rule for the class: before running any regression, plot X against Y in a scatterplot. If the points curve, cluster oddly, or contain obvious outliers, a straight line is the wrong tool — or you need to clean the data first. Code without a sanity-check plot is a common source of bad business decisions.

```
import matplotlib.pyplot as plt

plt.scatter(sqft, price)
plt.plot(sqft, [slope * x + intercept for x in sqft], color='red')
plt.xlabel('Square Footage')
plt.ylabel('Price (Rs Lakhs)')
plt.title('Always look at the plot before trusting the line!')
plt.show()
```

PART 3

Chi-Square Test — Testing Categorical Relationships

Everything so far — correlation, t-tests, regression — works on **numeric** data. But a huge amount of business data is **categorical**: yes/no, which plan did they choose, which region are they from. The **chi-square test of independence** is the tool for this.

The Question Chi-Square Answers

"Are these two categorical variables related, or are they independent?" For example: is there a relationship between *customer region* and *which subscription plan they choose*, or are those two things unrelated?

Null and Alternative Hypothesis (Same Framework as Thursday!)

	Statement
H0 (Null)	The two categorical variables are independent — no relationship
H1 (Alternative)	The two categorical variables are related — not independent

Talking point: emphasize this is the exact same hypothesis-testing logic from Thursday, just applied to a different kind of data. Students who understand p-values and alpha already have 80% of what they need for chi-square — only the test statistic itself is new.

Chi-Square — A Worked Example

Scenario: Does device type (Mobile vs. Desktop) relate to whether a customer completes checkout?

The Contingency Table (Observed Counts)

	Completed Checkout	Abandoned Checkout	Total
Mobile	120	180	300
Desktop	150	50	200
Total	270	230	500

Just by eyeballing this table: desktop users complete checkout at 75% (150/200), while mobile users complete it at only 40% (120/300). That's a big difference — but is it statistically significant, or could it be random noise in this particular sample of 500 customers? That's exactly what chi-square tells us.

Python: Run the Chi-Square Test

```
import numpy as np
from scipy.stats import chi2_contingency

# Rows = device type, Columns = [Completed, Abandoned]
observed = np.array([[120, 180],
                     [150, 50]])

chi2_stat, p_value, dof, expected = chi2_contingency(observed)

print(f"Chi-square statistic: {chi2_stat:.2f}")
print(f"P-value: {p_value:.6f}")
print(f"Degrees of freedom: {dof}")

if p_value < 0.05:
    print("Reject H0: device type and checkout completion ARE related")
else:
    print("Fail to reject H0: no significant relationship detected")
```


Chi-Square — Reading the Result

How the Test Actually Works (Plain English)

Chi-square compares your **observed** counts (what you actually saw) against the **expected** counts (what you'd expect if there were truly no relationship, i.e., if device type and checkout completion were independent). The bigger the gap between observed and expected, the larger the chi-square statistic, and the smaller the resulting p-value.

In This Example

The test returns a very small p-value (well below 0.05), so we reject H0. Conclusion: device type and checkout completion are **not independent** — there's a statistically significant relationship between which device someone uses and whether they finish checking out.

The Business Translation

This doesn't tell you *why* mobile checkout fails more often — remember correlation/association is not causation. But it gives the product team a clear, data-backed reason to investigate the mobile checkout flow specifically (Is the form too long? Are buttons too small? Is the page slow to load on mobile data?).

When to use chi-square vs. a t-test: use a t-test when comparing the *average of a numeric variable* across two groups (e.g., average order value, mobile vs. desktop). Use chi-square when comparing *counts or proportions of a categorical outcome* across groups (e.g., completed vs. abandoned checkout, mobile vs. desktop).

Which Test Should I Use?

Your Question	What Kind of Data	Use This Test
Do two numeric variables move together?	Both numeric	Correlation (Wednesday)
Can I predict one numeric variable from another?	Both numeric	Linear Regression (Today)
Is the average of a numeric variable different between two groups?	Numeric outcome, categorical group	T-Test (Thursday)
Is the average different across 3+ groups?	Numeric outcome, categorical group (3+ levels)	ANOVA (mentioned, not covered in depth)
Are two categorical variables related?	Both categorical	Chi-Square Test (Today)

Talking point: print or project this table and use it as the "master decision tree" for the week. Walk through a few rapid-fire examples and have students call out which test applies: "Does income predict spending?" (regression). "Do men and women prefer different payment methods?" (chi-square). "Is average spend higher in Region A than Region B?" (t-test).

AVOID THESE

Common Mistakes Students Make

Mistake 1: 'My R-squared is 0.95, so my model proves X causes Y.'

Reality: No. A high R-squared just means the line fits the data well — it says nothing about causation. Both could be driven by a hidden confounding variable, exactly like Wednesday's ice-cream-and-drowning example.

Mistake 2: 'I'll use my regression line to predict way outside my data's range.'

Reality: This is extrapolation, and it's risky. A line fit on 1000-2000 sqft homes tells you nothing reliable about a 10,000 sqft mansion.

Mistake 3: 'Chi-square works fine even with tiny sample sizes.'

Reality: Chi-square needs reasonably large expected counts in each cell (a common rule of thumb is at least 5) to be reliable. With very small samples, the test can give misleading p-values.

Mistake 4: 'I got a significant regression p-value, so the relationship is strong.'

Reality: Significance and strength are different. A large enough sample can make even a very weak relationship (low R-squared) statistically significant. Always check R-squared and the slope size, not just the p-value.

Mistake 5: 'Correlation, regression, and chi-square are basically interchangeable.'

Reality: They answer different questions on different data types. Use the decision guide on the previous page to pick the right tool for the data you actually have.

PRACTICE

Problems — Regression & Chi-Square

Problem 1: A regression of monthly ad spend on website visits gives slope = 12, intercept = 200, $R^2 = 0.81$. Interpret all three numbers in plain English.

Hint: each extra rupee of spend predicts ~12 more visits; with zero spend, ~200 visits are predicted anyway (organic traffic); 81% of the variation in visits is explained by ad spend.

Problem 2: You want to predict employee salary using years of experience, but your data only ranges from 0 to 15 years. Why is it risky to predict the salary of a 40-year veteran with this model?

Hint: this is extrapolation — predicting far outside the range of the original data, where the relationship may no longer be linear or may not hold at all.

Problem 3: A chi-square test on 'marketing channel' (Email, Social, Search) vs. 'made a purchase' (Yes, No) returns $p = 0.001$. What do you conclude?

Hint: reject H_0 — marketing channel and purchase outcome are significantly related (not independent); some channels likely convert better than others.

Problem 4: Your colleague wants to compare average revenue across 4 different regions. Should they use a t-test, chi-square, or something else?

Hint: with more than 2 groups and a numeric outcome, the right tool is ANOVA, not a t-test (t-tests compare exactly 2 groups) and not chi-square (that's for categorical outcomes).

Problem 5: Write the null and alternative hypotheses for: 'Is customer satisfaction rating (categorical: Low/Medium/High) related to which support channel (Phone/Chat/Email) the customer used?'

Hint: H_0 : satisfaction rating and support channel are independent. H_1 : satisfaction rating and support channel are related.

PART 4

From Statistics to the Job: SQL & Dashboards

All week we've worked with small, clean lists of numbers. In a real job, your data lives in a database, and your first job every day is pulling and shaping it before any statistics happen.

Where This Week's Concepts Show Up in SQL

Concept	Typical SQL / Dashboard Task
Correlation	Pull two numeric columns together with a JOIN, export to Python/Excel, run <code>.corr()</code>
Hypothesis Testing / A-B Testing	Query 'control' vs. 'treatment' user groups via a CASE WHEN or <code>experiment_id</code> column, compare metrics
Regression	Aggregate features (SUM, AVG, COUNT) per row as model inputs before predicting an outcome
Chi-Square	GROUP BY two categorical columns to build the contingency table directly in SQL
Dashboards	Tools like Looker, Tableau, or Power BI visualize these same metrics for non-technical stakeholders

Sample SQL: Building a Chi-Square Contingency Table

```
SELECT
  device_type,
  checkout_status,
  COUNT(*) AS customer_count
FROM checkout_events
GROUP BY device_type, checkout_status;
-- Export this result straight into pandas / scipy.stats.chi2_contingency()
```

Cheat Sheet — Friday

Concept	Definition	Key Point
Linear Regression	Best-fit straight line predicting Y from X	$Y = mX + b$; slope = change in Y per unit X
R-Squared	% of variation in Y explained by X	Equals r^2 for simple regression; higher = better fit
Extrapolation	Predicting outside your data's range	Risky — avoid unless you have strong reason to trust it
Chi-Square Test	Tests if two categorical variables are related	Same $H_0/H_1/p$ -value framework as Thursday's tests
Contingency Table	Cross-tab of counts for two categorical variables	The input data structure for chi-square
Decision Guide	Numeric+numeric = correlation/regression; numeric+categorical(2) = t-test; categorical+categorical = chi-square	Pick the test based on data types, not habit

Key Takeaways & The Week in Review

Key Takeaways from Friday

- Regression turns correlation into a predictive equation: $Y = mX + b$
- R-squared (which equals r^2) tells you how much of Y's variation X explains — not causation
- Watch for extrapolation, outliers, and non-linear patterns before trusting a regression line
- Chi-square extends hypothesis testing to categorical data using the same H_0/H_1 /p-value logic from Thursday
- Use the decision guide: data types (numeric vs. categorical) determine which test is appropriate
- Every concept this week eventually gets expressed as a SQL query or a dashboard metric on the job

The Full Week, in One Table

Day	Big Idea
Tuesday	Probability and the normal distribution describe uncertainty and what's 'typical'
Wednesday	Correlation shows direction and strength of a relationship between numeric variables
Thursday	Hypothesis testing tells us if an observed difference is real or just chance
Friday	Regression predicts outcomes; chi-square tests relationships in categorical data

Congratulations: you now have the core statistical toolkit that underlies almost every data analyst task — from exploratory analysis, to A/B testing, to building predictive models, to presenting clean, defensible findings to stakeholders.